

SpaceX Falcon 9

First Stage Landing Prediction



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PLATFORM: IBM Skills Network / Coursera

GitHub: <https://github.com/Tulsi-Naik/spacex-capstone>

TOOLS: Python · SQL · Folium · Plotly Dash · scikit-learn

Executive Summary

OVERVIEW

90+

Total Launches

4

Launch Sites

4

ML Models Built

83.3%

Best Accuracy



Project Goals

- Predict Falcon 9 first-stage landing success to estimate launch cost savings
- SpaceX charges \$62M vs. competitors' \$165M+; reuse is the key differentiator
- Enable rival bidders to forecast launch costs based on landing probability
- Build end-to-end data science pipeline: collection → EDA → ML prediction



Key Findings

- KSC LC-39A had the highest launch success rate among all four sites
- Success rate improves with increasing flight number — experience matters
- Orbits ES-L1, GEO, HEO, SSO achieved 100% success rates
- Decision Tree & SVM both achieved 83.3% accuracy on hold-out test set
- F9 B5 booster showed strongest landing success correlation

Introduction

BACKGROUND

Problem Statement

Can we predict whether the Falcon 9 first stage will land successfully, enabling cost-competitive launch pricing analysis for alternative providers bidding against SpaceX?



Dataset Overview

- SpaceX REST API — rocket, payload, launchpad metadata
- Wikipedia web scrape — historical Falcon 9 launch records
- SpaceX.csv — 90+ Falcon 9/Heavy missions
- Fields: Date, Launch Site, Payload Mass, Orbit, Booster Version, Landing Outcome



Project Objectives

- Collect data via SpaceX API & web scraping
- Perform SQL + visual EDA to identify patterns
- Build interactive Folium maps of launch sites
- Develop Plotly Dash analytical dashboard
- Train & compare ML classification models



Technologies Used

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL / SQLite — structured data queries
- Folium — interactive geographic mapping
- Plotly Dash — interactive web dashboard
- scikit-learn — ML model training & evaluation

Data Collection & Wrangling Methodology

LAB 1 & 2

SpaceX API
GET Request

Wikipedia
Web Scraping

Data
Cleaning

Feature
Engineering

Training
Labels



Data Collection Sources

- SpaceX REST API v4: rockets, launchpads, payloads, cores
- Helper functions: getBoosterVersion(), getLaunchSite(), getPayloadData(), getCoreData()
- HTTP GET requests via Python requests library
- Wikipedia: List of Falcon 9 & Falcon Heavy Launches
- BeautifulSoup4 HTML parser to extract launch tables
- Structured into Pandas DataFrame (90 rows × 17 features)



Wrangling & Preprocessing

- Null value analysis: PayloadMass, LandingPad had missing values — imputed with column mean
- Landing outcomes encoded: True/False × Ocean/RTLS/ASDS → binary Class (1=success, 0=failure)
- Identified 4 launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
- Orbit types classified: LEO, GTO, ISS, SSO, VLEO, MEO, GEO, HEO, ES-L1, PO
- StandardScaler applied for ML feature normalization
- 80/20 train-test split with random_state=2

EDA & Interactive Visual Analytics Methodology

LABS 3–5



EDA with Matplotlib & Seaborn

- catplot: FlightNumber vs LaunchSite, PayloadMass vs LaunchSite (hue=Class)
- Bar charts: success rate per orbit type
- Line chart: yearly launch success trend
- Feature correlation heatmap for ML feature selection



SQL-Based EDA (SQLite)

- Loaded SpaceX CSV into SQLite SPACEXTBL via ipython-sql
- Aggregation queries: COUNT, SUM, AVG, MIN, MAX
- Filtering with LIKE, WHERE conditions
- GROUP BY for outcome distributions & year analysis



Folium Interactive Maps

- Mapped all 4 launch sites with CircleMarker (color = success/fail)
- MarkerCluster plugin for dense launch point grouping
- MousePosition plugin for real-time lat/lon display
- PolyLine overlays for proximity distances to coast, rail, highway



Plotly Dash Dashboard

- Dropdown: filter by launch site (All / individual)
- RangeSlider: filter Payload Mass 0–10,000 kg
- Pie chart callback: success distribution per site
- Scatter chart callback: payload vs. outcome colored by Booster Version

Predictive Analysis Methodology

LAB 5 — ML

1. Load
dataset_part_2
& part_3

2. Extract
Class column
as Y array

3. StandardScaler
normalize X

4. train_test_split
80/20 split
random_state=2

5. GridSearchCV
cv=10 folds
Hyperparameter

6. Compare
4 Classifiers
& Select Best

Logistic Regression

$C \in \{0.01, 0.1, 1\}$
penalty: L2 (Ridge)
solver: lbfgs

Support Vector Machine

kernel: linear/rbf/poly/sigmoid
C: logspace(-3,3,5)
gamma: logspace(-3,3,5)

Decision Tree Classifier

criterion: gini/entropy
max_depth: 2-18
min_samples_leaf: 1/2/4

K-Nearest Neighbors

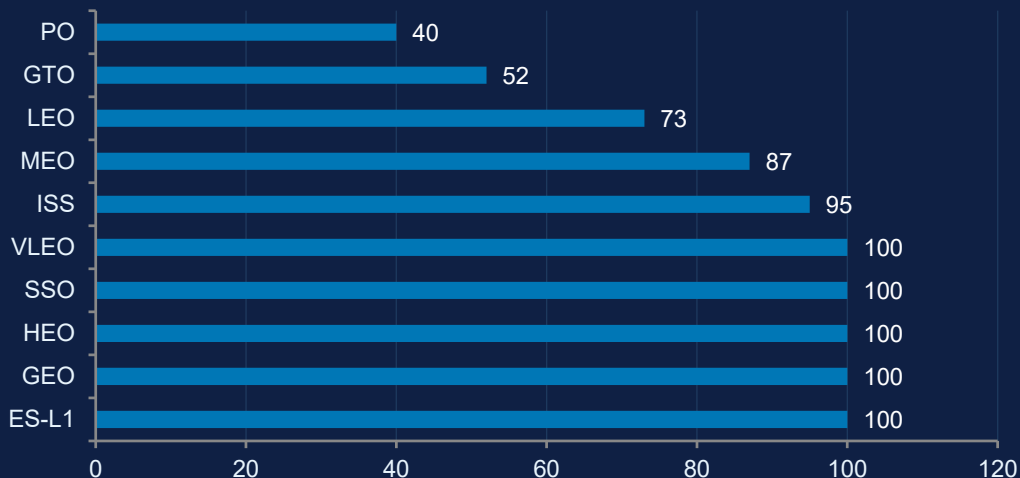
n_neighbors: 1-10
algorithm:
auto/ball_tree/kd_tree/brute
p: 1 or 2 (Manhattan/Euclidean)

Evaluation: Accuracy score on 18 test samples + Confusion Matrix (seaborn heatmap)

EDA Visualization Results

LAB 3 RESULTS

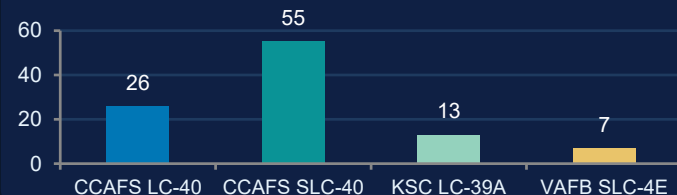
Success Rate by Orbit Type (%)



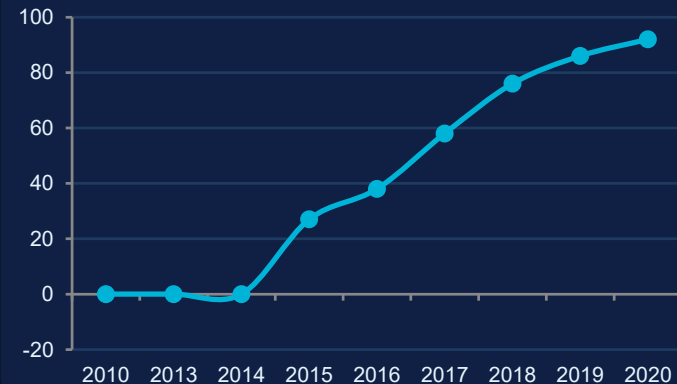
Key Findings

- ES-L1, GEO, HEO, SSO, VLEO orbits achieved 100% success — highest-value missions
- Flight number strongly correlates with landing success — booster experience is crucial
- VAFB SLC-4E had zero heavy-payload (>10,000 kg) launches in dataset

Launches per Site



Yearly Landing Success Rate (%)



EDA with SQL Results

LAB 4 — SQL

Task 1 — Unique Launch Sites

```
SELECT DISTINCT "Launch_Site" FROM SPACEXTBL;
```

→ CCAFS LC-40 · CCAFS SLC-40 · KSC LC-39A · VAFB SLC-4E

Task 3 — Total NASA (CRS) Payload Mass

```
SELECT SUM("PAYLOAD_MASS__KG_") AS Total_Payload_Mass  
FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

→ 45,596 kg total payload carried for NASA CRS missions

Task 5 — First Successful Ground Pad Landing

```
SELECT MIN(Date) AS First_Successful_Landing  
FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (ground pad)';
```

→ 2015-12-22 — First successful ground pad landing achieved

Task 6 — Boosters: Drone Ship + 4000–6000 kg Payload

```
SELECT DISTINCT "Booster_Version" FROM SPACEXTBL  
WHERE "Landing_Outcome" = 'Success (drone ship)'  
AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;
```

→ F9 FT B1022 · F9 FT B1026 · F9 FT B1021.2 · F9 FT B1031.2

Task 7 — Mission Outcome Counts

```
SELECT "Mission_Outcome", COUNT(*) AS Outcome_Count  
FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

→ Success: 98 · Failure (in flight): 1 · Success (payload status unclear): 1

Task 8 — Boosters with Max Payload Mass

```
SELECT DISTINCT "Booster_Version" FROM SPACEXTBL  
WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_")  
FROM SPACEXTBL);
```

→ F9 B5 B1048.4 · F9 B5 B1049.4 · F9 B5 B1051.3 · F9 B5 B1056.4 · F9 B5 B1048.5

Interactive Map Results — Folium

LAB 5 — MAPS



SpaceX Launch Site Coordinates

Launch Site	Latitude	Longitude	Location
CCAFS LC-40	28.5621°N	80.5774°W	Cape Canaveral, FL
CCAFS SLC-40	28.5620°N	80.5770°W	Cape Canaveral, FL
KSC LC-39A	28.6080°N	80.6040°W	Kennedy Space Center, FL
VAFB SLC-4E	34.6325°N	120.6108°W	Vandenberg AFB, CA

- MarkerCluster — groups launches geographically at varying zoom levels
- CircleMarker — green=success, red=failure, radius proportional to payload
- MousePosition plugin — live lat/lon readout on cursor hover
- PolyLine overlays — measured proximity distances to nearest coastline, railway, and highway
- DivIcon labels — site name annotations directly on the map

Geographic Insights

- All US launch sites cluster near coastlines — ocean safety corridor for booster descent
- Cape Canaveral sites (FL) benefit from Earth's equatorial rotation speed boost
- KSC LC-39A proximity to rail and road infrastructure enables rapid payload integration
- VAFB SLC-4E (CA) favors polar & SSO orbits due to southward trajectory over Pacific
- No sites located inland — ocean proximity is a hard safety and operational constraint
- Minimum distance to coastline < 1 km for all four active SpaceX sites

Plotly Dash Dashboard Results

LAB 6 — DASHBOARD

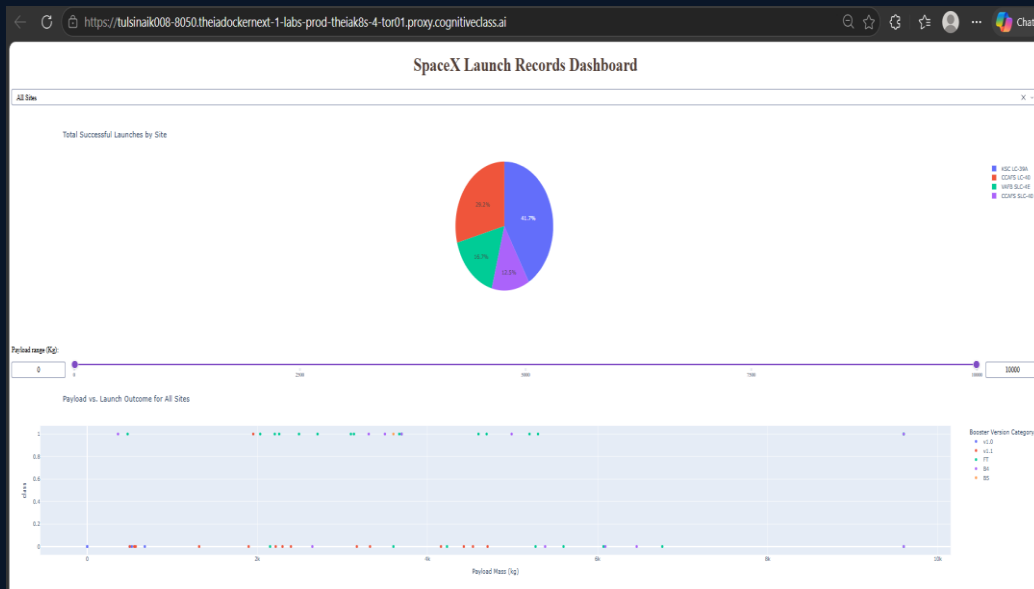


Fig 1: Full dashboard — All Sites, payload 0–10,000 kg



Fig 2: KSC LC-39A selected — payload with range between 2000 -6000

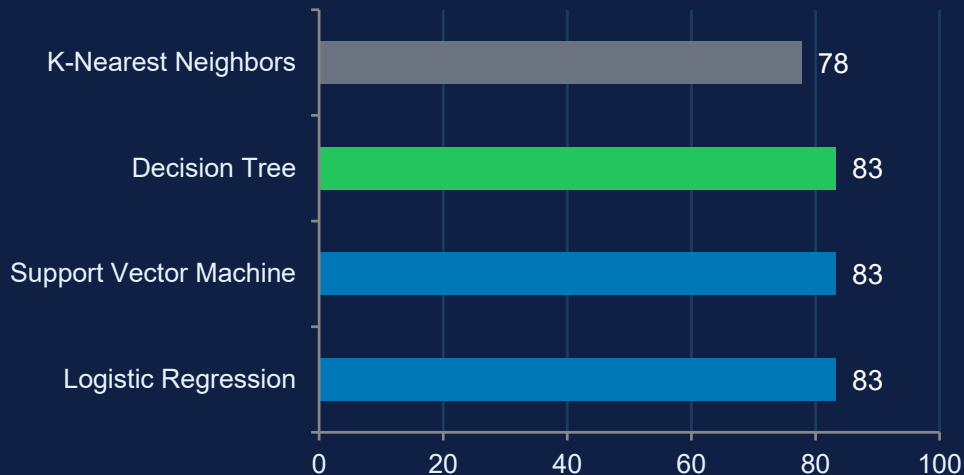
Dashboard Components

- Dropdown: All Sites / CCAFS LC-40 / CCAFS SLC-40 / KSC LC-39A / VAFB SLC-4E
- Pie chart: total success by site (All) or success/fail ratio (per site)
- RangeSlider: 0–10,000 kg payload mass filter with 1,000 kg step
- Scatter: Payload Mass vs Class, colored by Booster Version Category

Predictive Analysis Results

LAB 5 — ML

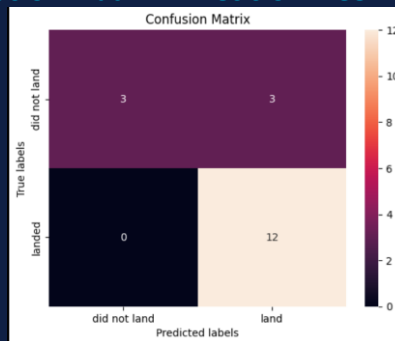
Model Test Accuracy (%)



Best Model: Decision Tree (tied)

All three top models achieved 83.33% accuracy (15/18 correct). Decision Tree selected for interpretability.

Confusion Matrix — Decision Tree



- True Positives (TP): 12 — landed correctly predicted
- True Negatives (TN): 3 — no-land correctly predicted
- False Positives (FP): 3 — predicted land, did not land
- False Negatives (FN): 0 — no missed landing predictions
- Precision: 80.0% | Recall: 100% | F1: 0.889
- KNN scored 77.78% — lowest due to 4 misclassifications

Conclusion

SUMMARY



Project Delivered

- Full end-to-end data science pipeline built from scratch
- Data collected via API, web scraping, and CSV ingestion
- SQL EDA, visual EDA, Folium maps, and Dash dashboard all completed
- 4 ML classifiers trained, tuned, and evaluated with GridSearchCV



Key Learnings

- Booster reuse experience (flight number) is the strongest success predictor
- Launch site geography directly constrains orbit types achievable
- ES-L1, GEO, HEO missions all achieved 100% landing success
- KSC LC-39A leads in success rate among all four active sites



Model Performance

- Best test accuracy: 83.33% (Logistic Regression, SVM, Decision Tree)
- Decision Tree preferred for interpretability and feature visibility
- Small test set (18 samples) limits statistical confidence — noted
- All models outperform 50% baseline, confirming predictive signal



Future Improvements

- Expand dataset with 2021–2024 Starship and Falcon 9 flights
- Engineer weather and sea-state features at launch time
- Explore ensemble methods (Random Forest, XGBoost) for higher accuracy
- Deploy model as REST API to enable real-time cost estimation tool

This capstone demonstrates a complete data science workflow: collection → wrangling → EDA → visualization → predictive modeling